In [ ]:
```
import pandas as pd
import numpy as np

data = {
    "Age": [25, 30, np.nan, 28, 45, 50, 120],
    "Salary": [30000, 45000, 50000, np.nan, 70000, 80000, 1000000],
    "Gender": ["Male", "Female", "Male", "Female", np.nan, "Male", "Female"],
    "City": ["Mumbai", "Pune", "Mumbai", "Delhi", "Pune", "Delhi", "Mumbai"]
}

df = pd.DataFrame(data)
df.head()
```

Out[ ]:

	Age	Salary	Gender	City
0	25.0	30000.0	Male	Mumbai
1	30.0	45000.0	Female	Pune
2	NaN	50000.0	Male	Mumbai
3	28.0	NaN	Female	Delhi
4	45.0	70000.0	NaN	Pune

In [ ]:
```
print(df.isnull().sum())
```

```
Age       1
Salary    1
Gender    1
City      0
dtype: int64
```

In [ ]:
```
from sklearn.impute import SimpleImputer
numeric_cols = df.select_dtypes(include=np.number).columns
num_imputer = SimpleImputer(strategy='mean')
df[numeric_cols] = num_imputer.fit_transform(df[numeric_cols])
categorical_cols = df.select_dtypes(include='object').columns
cat_imputer = SimpleImputer(strategy='most_frequent')
df[categorical_cols] = cat_imputer.fit_transform(df[categorical_cols])
print(df)
```

```
          Age     Salary  Gender    City
0   25.000000    30000.0    Male  Mumbai
1   30.000000    45000.0  Female    Pune
2   49.666667    50000.0    Male  Mumbai
3   28.000000   212500.0  Female   Delhi
4   45.000000    70000.0  Female    Pune
5   50.000000    80000.0    Male   Delhi
6  120.000000  1000000.0  Female  Mumbai
```

In [ ]:
```
df_encoded = pd.get_dummies(df)
print(df_encoded)
```

```
          Age     Salary  Gender_Female  Gender_Male  City_Delhi  City_Mumbai  \
0   25.000000    30000.0          False         True       False         True
1   30.000000    45000.0           True        False       False        False
2   49.666667    50000.0          False         True       False         True
3   28.000000   212500.0           True        False        True        False
4   45.000000    70000.0           True        False       False        False
5   50.000000    80000.0          False         True        True        False
6  120.000000  1000000.0           True        False       False         True

   City_Pune
0      False
1       True
2      False
3      False
4       True
5      False
6      False
```

In [ ]:
```
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df_scaled = df_encoded.copy()
df_scaled[numeric_cols] = scaler.fit_transform(df_scaled[numeric_cols])
print(df_scaled)
```

```
        Age    Salary  Gender_Female  Gender_Male  City_Delhi  City_Mumbai  \
0  0.000000  0.000000          False         True       False         True
1  0.052632  0.015464           True        False       False        False
2  0.259649  0.020619          False         True       False         True
3  0.031579  0.188144           True        False        True        False
4  0.210526  0.041237           True        False       False        False
5  0.263158  0.051546          False         True        True        False
6  1.000000  1.000000           True        False       False         True

   City_Pune
0      False
1       True
2      False
3      False
4       True
5      False
6      False
```

In [ ]:
```
from sklearn.preprocessing import StandardScaler
std = StandardScaler()
df_standard = df_encoded.copy()
df_standard[numeric_cols] = std.fit_transform(df_standard[numeric_cols])
print(df_standard)
```

```
        Age    Salary  Gender_Female  Gender_Male  City_Delhi  City_Mumbai  \
0 -0.813657 -0.559119          False         True       False         True
1 -0.648727 -0.513164           True        False       False        False
2  0.000000 -0.497846          False         True       False         True
3 -0.714699  0.000000           True        False        True        False
4 -0.153935 -0.436572           True        False       False        False
5  0.010995 -0.405936          False         True        True        False
6  2.320022  2.412637           True        False       False         True

   City_Pune
0      False
1       True
2      False
3      False
4       True
5      False
6      False
```

In [ ]:
```
for col in numeric_cols:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[
        (df[col] < lower) | (df[col] > upper)
    ]

    print(col)
    print(outliers)
```

```
Age
     Age     Salary  Gender    City
6  120.0  1000000.0  Female  Mumbai
Salary
     Age     Salary  Gender    City
6  120.0  1000000.0  Female  Mumbai
```

In [ ]:
```
df_clean = df.copy()
```

In [ ]:
```
for col in numeric_cols:

    Q1 = df_clean[col].quantile(0.25)
    Q3 = df_clean[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df_clean = df_clean[
        (df_clean[col] >= lower) &
```

```
        (df_clean[col] <= upper)
    ]
```

In [ ]:
```
print(df_clean)
```

```
         Age   Salary  Gender    City
0  25.000000  30000.0    Male  Mumbai
1  30.000000  45000.0  Female    Pune
2  49.666667  50000.0    Male  Mumbai
4  45.000000  70000.0  Female    Pune
5  50.000000  80000.0    Male   Delhi
```

In [ ]:
```
!jupyter nbconvert --to html /content/smld_9_new.ipynb
```